

Answer Key

Question 1

Both portfolios offer the same average return (9%), but Portfolio B has more than double the standard deviation of Portfolio A (18% vs 8%). Since standard deviation measures how widely returns fluctuate around the average, Portfolio B is significantly riskier without offering any additional return to compensate for that risk.

A risk-averse investor would prefer Portfolio A. For the same expected return, a rational investor always prefers less volatility, since it means more predictable outcomes and lower chances of a severe loss in any given period.

Question 2

Expected change = Beta $\times$ Market change

Expected change = $1.3 \times 10\% = 13\%$

A beta of 1.3 means the portfolio is expected to move 1.3 times as much as the overall market. If the market rises 10%, the portfolio should rise by approximately 13%. (Note: this also means the portfolio would be expected to fall by about 13% if the market fell 10%, since beta cuts both ways.)

Question 3

Holding stocks across multiple industries reduces unsystematic (firm- or industry-specific) risk. This type of risk comes from events that affect one company or sector but not the broader economy, such as a product recall, a labour strike, or an industry-specific regulatory change.

Because these events are largely independent of one another, a loss in one holding (say, an airline hurt by rising fuel costs) is likely to be offset, or at least not mirrored, by holdings in unrelated industries (such as healthcare or technology). As more uncorrelated holdings are added, this firm-specific noise increasingly cancels out, and total portfolio risk falls.

This diversification benefit has a limit, though: it only reduces unsystematic risk. Systematic (market-wide) risk, such as a recession or an interest rate shock, affects all industries simultaneously and cannot be diversified away.